

Motivating R&D team creativity: the social cognitive effect of external developmental feedback and internal goal interdependence

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The relationship between external communication and R&D team creativity has received increasing attention in the R&D management literature. However, little is known about the motivational mechanism between the two. We propose a second-stage moderation model based on social cognitive theory to explain whether, how and when R&D team's external developmental feedback motivates its creativity. Using the data gathered from 452 R&D team members and 84 supervisors in a time-lagged field study, we found external developmental feedback had a positive effect on R&D team creativity via increasing team creative efficacy. Further, conditional indirect effect analyses found that a high level of cooperative goal interdependence strengthened the positive effect of team creative efficacy on R&D team creativity and the mediating effect of team creative efficacy between external developmental feedback and R&D team creativity.

1. Introduction

Nowadays R&D team creativity, which refers to the collective generation of novel and useful ideas by R&D team members, has been suggested as an important source of a firm's competitiveness (Chen et al., 2010). The generation of creative ideas as a team requires members' collective confidence and cooperation (Shin and Eom, 2014), hence managers

should integrate external and internal resources to motivate team members. In the existing research exploring the motivation of R&D team creativity (e.g. Shin and Zhou, 2007; Boon et al., 2016; Wang et al., 2016), a social cognitive mechanism has received increased attention (Bandura, 1982), which suggests social factors could motivate R&D team creativity via increasing team creativity efficacy — a kind of collective efficacy defined as 'team members' shared

belief in the team's capability to generate creative ideas' (Shin and Zhou, 2007, p. 1138).

R&D teams often actively interact with external experts, clients, funding sources and other stakeholders outside the organization and seek feedback from them (Tang, 2016). Among feedback received by the R&D team from external communications, some are developmental feedback which are composed of useful information and could direct the R&D team to make progress (Zhou, 2003). Drawing on social cognitive theory (Bandura, 1986), external developmental feedback may motivate R&D team creativity by providing important evidence for R&D team members' judgment of team creative efficacy. Despite the growing empirical evidence shows the positive impact of external communications on the R&D team's innovation or creativity (e.g. Tang, 2014; Tang, 2016; Hoever et al., 2018), the existing studies mainly use the access and processing of information or knowledge to explain the influencing mechanism. Little research has examined whether, how and when R&D team's external communications may promote its creativity from a motivational perspective. Therefore, the purpose of this study is to fill the void by developing and examining a theoretical framework about the motivational effect of external developmental feedback on R&D team creativity via increasing team creative efficacy on the basis of social cognitive theory. Our results may add to the R&D team management literature by elucidating the underlying motivational mechanism whereby external communication drives R&D team creativity.

Although Bandura's social cognitive theory has noted that the motivational effect of collective efficacy on group outputs is contingent on the level of group members' collective action based on cooperative goals (Bandura, 1986), prior research on the motivation of R&D team creativity has not yet included the boundary condition (moderator) of the relationship between team creative efficacy and team creativity in their framework (e.g. Shin and Zhou, 2007; Yang et al., 2017). As a result, there is no substantive empirical evidence to determine when social factors could motivate R&D team creativity via increasing team creative efficacy. Drawing on the theory of cooperation and competition, as one of the most important prerequisites of team members' collective actions (Deshon et al., 2004; Tjosvold et al., 2004; De Dreu, 2007), cooperative goal interdependence refers to the extent to which team members believe that their individual goal achievement is conditional on the achievement of team goals and other members' goals (Tjosvold et al., 2004). In this study, we integrated social cognitive theory and the theory of cooperation and competition to expose how

cooperative goal interdependence may amplify the influence of team creative efficacy on R&D team creativity. The results of this study could also help to identify the boundary condition that qualifies the motivational effect of external developmental feedback on R&D team creativity via team creative efficacy. In sum, the current study developed an integrated theoretical model in order to fully capture the external and internal dynamics at play in motivating R&D team creativity from a social cognitive perspective.

2. Theoretical background and hypotheses

2.1. *The motivation of R&D team creativity from a social cognitive perspective*

Early researches reveal the formation mechanism of R&D team's creativity mainly from the cognitive, informational and emotional perspectives (e.g. Leenders et al., 2003; Kratzer et al., 2008). A high level of R&D team creativity actually requires motivating team members to set the challenging goal, cooperate with each other and continue their creative endeavours when facing obstacles (Garcia Martinez et al., 2017). Although still limited in amount, recent research has emerged to explore the motivation of R&D team creativity (e.g. Shin and Zhou, 2007; Boon et al., 2016; Wang et al., 2016). The most influential ones draw on Bandura's social cognitive theory which emphasizes the critical role of social factors in motivating group behaviour and outputs through increasing collective efficacy (Bandura, 1982).

Social cognitive theory advocates collective efficacy, defined as a group's belief that 'they can solve their problems and improve their lives through concerted effort' (Bandura, 1982, p. 143), as a key motivator for a group in setting challenging goals and sustaining their efforts in face of difficulties (Bandura, 1982). Building on social cognitive theory, Shin and Zhou (2007) developed the team creative efficacy scale. They argued that the conceptually narrower team creative efficacy might be more relevant to creativity than general collective efficacy because teams with higher creative efficacy are more likely to break away from traditional ways of doing things, set a challenging goal of generating creative ideas and persevere in the face of difficulties (Shin and Zhou, 2007). Using social cognitive theory for understanding how work teams and particularly R&D teams are motivated to be creative in a social context, researchers highlight team creative efficacy as the key mechanism transmitting the impact of contextual factors

within the team to team creativity (e.g. Shin and Zhou, 2007; Hon and Chan, 2013; Yang et al., 2017). Extending the literature on the motivation of R&D team creativity from a social cognitive perspective, we contend that the R&D team's external communication could motivate its creativity by offering external information cues for the development of team creative efficacy.

2.2. Social cognitive effect of external developmental feedback on R&D team creativity

2.2.1. External developmental feedback

In order to keep up with the rapid development of technology and knowledge, most R&D team leaders have to frequently seek feedback from external experts, clients, funding sources, or other stakeholders outside the organization about the team's development. As R&D generally requires professional advice, R&D teams increasingly highlight the quality rather than the quantity of external communication in order to have more chances to receive external developmental feedback which is informational in nature and focuses on R&D teams' development and improvement (Chatenier et al., 2010). When external experts or other stakeholders provide useful information for the development of the whole R&D team, intra-team communication about the details of external developmental feedback will be carried out and team members' perception about the extent to which the team receives external developmental feedback emerges into the team level (Deshon et al., 2004). Accordingly, we focus on external developmental feedback targeted at the R&D team as a whole, and regard R&D teams' external experts, clients, funding sources, or other stakeholders outside the organization as sources of developmental feedback. In this study, external developmental feedback is defined as the extent to which external stakeholders provide teams with helpful and useful information that enables members to learn, develop, and make improvements on the team task, and measured as R&D team members' collective perception about team-level feedback.

2.2.2. External developmental feedback and team creative efficacy

According to social cognitive theory, there are four primary sources of information based on which people develop their beliefs about efficacy: social persuasion, physiological state, vicarious experience and mastery experience (Bandura, 1982). Extending theory to the R&D team context, we argue that external developmental feedback can cultivate these

sources of information and improve team creative efficacy. First, when providing external developmental feedback, external experts or other stakeholders are more likely to encourage the R&D team to take on challenging task assignments (Joo et al., 2012), which acts as effective social persuasion shaping the R&D team's creative efficacy (Bandura, 1982). Second, developmental feedback from external ties may accord the R&D team a supportive network (Tang, 2016), which will help team members reduce aversive physiological arousal such as stress and anxiety while accomplishing challenging tasks (Liao et al., 2010). The enhanced social-emotional support may improve team members' shared creative efficacy belief (Bandura, 1982; Zhou, 2003). Thirdly, when external experts or other stakeholders give R&D team useful information for the progress of R&D task, team members may be inclined to choose those feedback providers as social models to observe and learn from, which will contribute to their shared perception of the team's creative capabilities (Shin and Zhou, 2007). Finally, given the useful information, encouragement, and high expectations provided as external developmental feedback to the R&D team, and the R&D team will gradually accumulate experiences in generating creative ideas. Such mastery experiences of creative idea generation yield the most influential source of information about team creative efficacy (Bandura, 1982). Hence, we predict the following:

Hypothesis 1 External developmental feedback improves team creative efficacy.

2.2.3. External developmental feedback, team creative efficacy and team creativity

Social cognitive theory not only provides a detailed account of how collective efficacy is malleable in a social context but also sheds light on how it further plays a central role in a group regarding performing collective, risky and challenging tasks (Bandura, 1982; Bandura, 1986). According to the theory, a team's creative efficacy influences the amount of effort the team is willing to exert and how long they are willing to persevere when collective efforts fail to generate creative ideas (Bandura, 1982). When team creative efficacy is increased by external developmental feedback, R&D team members believe that their endeavours toward creativity will not be wasted (Shin and Zhou, 2007). They are more willing to take risks in trying different perspectives, procedures and possibilities to produce creative ideas that further fuel team creativity and will persist in generating creative ideas to solve the problem in front of failures and difficulties (Hon

and Chan, 2013; Yang et al., 2017). In this way, a high level of team creativity is achieved. Empirical evidence has indicated that team creative efficacy plays a pivotal role in the enhancement of team creativity and is a crucial mediating mechanism through which contextual factors motivate team creativity (e.g. Shin and Zhou, 2007; Hon and Chan, 2013; Yang et al., 2017). Thus, we predict the following:

Hypothesis 2 External developmental feedback improves R&D team creativity through its positive effect on team creative efficacy.

2.3. *The moderating role of R&D team's cooperative goal interdependence*

Although social cognitive theory demonstrates that collective efficacy plays a role in motivating group members to generate desired group outputs (Bandura, 1982), it also emphasizes the importance of members' cooperation in the motivational process. Bandura (1986) noted that when group members can't take collective action, even with a high level of collective efficiency, they will not be able to overcome the difficulties in achieving group goals.

The theory of cooperation and competition suggests that team members perceive their goals to be cooperatively or competitively linked (Deutsch, 1949). As a structural property of teams, cooperative goal interdependence represents collaboration for goal achievement in the team (Tjosvold et al., 2004). With a high level of cooperative goal interdependence, team members are able to take collective action to promote mutual goals (De Dreu, 2007). Integrating social cognitive theory and the theory of cooperation and competition, we propose that whether R&D team members could take collective action on the basis of cooperative goal interdependence is a key condition for the transition from team creative efficacy to R&D team creativity (Deutsch, 1949; Bandura, 1986).

With a high level of cooperative goal interdependence, R&D team members perceive that they 'swim or sink together' (De Dreu, 2007, p. 628). The interpersonal helping, coordination and comprehensive sharing of valuable resources are encouraged in the R&D team (Tjosvold et al., 2004; Chen et al., 2010), which in turn makes team members with a shared belief of high team creative efficacy cooperate with each other to generate creative ideas (Bandura, 1986), increasing the strength of the positive effect of team creative efficacy on R&D team creativity. In contrast, with a low level of cooperative goal interdependence, team member's individual goal achievement is competitive with the achievement of team

goals (Tjosvold et al., 2004). Even if the members have confidence in the R&D team's capability to be creative, they won't share important resources (e.g. data, information and knowledge) with each other or devote too much energy to the collective generation of creative outputs (De Dreu, 2007). Then R&D team's creative idea generation could not be effectively motivated by high team creative efficacy. Thus, we propose the following:

Hypothesis 3 R&D team's cooperative goal interdependence moderates the positive relationship between team creative efficacy and R&D team creativity: The positive relationship is stronger when cooperative goal interdependence is higher.

As articulated in the previous section on the basis of social cognitive theory, external developmental feedback enhances R&D team creativity through team creative efficacy. Integrating the theory of cooperation and competition into social cognitive theory, cooperative goal interdependence can enhance the influence of team creative efficacy on R&D team creativity by shaping cooperative behavioural patterns (Deutsch, 1949). Taking these arguments together, external developmental feedback has a positive impact on team creative efficacy, and cooperative goal interdependence exerts a positive moderating effect for the subsequent transition from team creative efficacy to R&D team creativity. When the path between the mediator and the dependent variable of a mediation model is theorized to be moderated by another variable, that moderating effect is referred to as the second-stage moderating effect (Edwards and Lambert, 2007). Hence, we expect cooperative goal interdependence to have a second-stage moderating effect on the influence of external developmental feedback on R&D team creativity via team creative efficacy.

Hypothesis 4 The indirect effect of external developmental feedback on R&D team creativity via team creative efficacy is moderated by the R&D team's cooperative goal interdependence, such that the indirect effect is more positive when cooperative goal interdependence is higher.

The conceptual model of this study is shown in Figure 1.

3. Methods

3.1. *Setting, participants, and procedures*

We used a survey questionnaire to collect the data and ensured the clarity of each item through a pilot

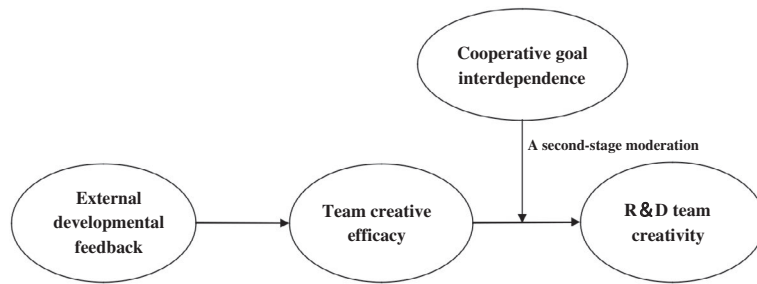


Figure 1. The conceptual model.

study on 10 R&D teams selected from one IT manufacturer in Xi'an, Shaanxi Province, China. Then we contacted and visited top managers of 32 high-tech companies in Shaanxi Province of China to discuss the study's purpose and invite them to introduce their R&D team leaders. 4 managers refused to participate because of the heavy work overload of their R&D teams. Then we contacted the R&D team supervisors in the 28 participating companies with an explanation of the objectives and requirements of this survey and a promise of confidentiality and feedback in return. 118 supervisors responded positively and gave us the name lists of their team members. All 829 participants in the 118 teams received a questionnaire and an e-mail to invite them to participate in the study and ensure the individual anonymity.

To avoid common method biases (Podsakoff et al., 2003), we collected data on different variables at different time points and from multiple sources. The questionnaire administered to team members in the first-wave survey contained items assessing perceived external developmental feedback and demographic information. A total of 614 employees in 102 R&D teams completed the first-wave survey. Two months later, 614 team members participating in our first-wave survey were asked to evaluate team creative efficacy and cooperative goal interdependence. A total of 461 employees in 85 R&D teams completed the second-wave survey. One month later, 85 R&D team supervisors were asked to rate the survey evaluating R&D team creativity. Only 84 supervisors finished the final evaluation because 1 supervisor was transferred during the survey.

To reduce potential biases when individual responses were aggregated to the team level, we used a 60% response rate as the criterion for including teams in the analyses (Somech and Drach-Zahavy, 2013), resulting in a final sample of 84 teams from 16 high-tech companies engaged in the development of electronic products (23 teams), machinery and equipment (15 teams), medical and pharmaceutical products (17 teams), application software (21

teams) and others (8 teams), consisting of 452 team members and 84 supervisors, with 3 to 14 respondents per team ($M = 5.4$ members). All R&D teams performed nonroutine, complex tasks that required differential expertise and skills. Most of the R&D teams have regular interactions with external experts and other stakeholders from whom they can receive developmental feedback. Through socialization and interactions at the team level, team members could develop a shared perception of external developmental feedback. The size of the complete team ranges between 3 and 16 ($M = 7$ members). *T*-test indicated no significant differences between respondents and non-respondents with respect to gender, age and tenure (all P 's $> .10$). The average age of the team supervisors was 41.5 years ($SD = 8.5$) and 80.3% of the leaders were male. The average age of the team members was 34.8 years ($SD = 7.4$), and 63.6% of participating team members were male. Team members had received higher education, that was, either a technical (23%) or a university (77%) degree.

3.2. Measures

As team members were Chinese speaking, all materials were presented in Chinese and most of the measurement items for the constructs were adapted with back-translation from well-established instruments in previous studies with similar research settings as our study. Responses were obtained on a Likert-type scale ranging from 'not at all correct' (1) to 'completely correct' (7).

3.2.1. External developmental feedback

We developed our measuring scale about external developmental feedback based on Zhou's (2003) three-item scale. Adjustments have been made to make the items applicable to the research background of the R&D team. For example, Zhou's (2003) original individual-level scale was adapted into team-level scale by changing 'I' into 'My team' (Deshon et al., 2004). Sample item was 'While giving my team feedback, external clients, experts, funding sources,

or other external stakeholders focus on helping the team to learn and improve’.

3.2.2. Team creative efficacy

Team creative efficacy was measured with three items from Shin and Zhou’s (2007) scale. Shin and Zhou (2007) used the scale to measure R&D team’s creative efficacy in Korean companies. Sample item was ‘My team has confidence in its ability to solve problems creatively’.

3.2.3. Cooperative goal interdependence

Five-item scale adapted from Tjosvold et al.’s (2004) research was used to measure cooperative goal interdependence. Tjosvold et al., (2004) used the scale to measure the work team’s cooperative goal interdependence in Chinese companies. Sample item was ‘When team members work together, we usually have common goals’.

3.2.4. R&D team creativity

The R&D team creativity scale with four items was adapted from Shin and Zhou’s (2007) scale, and rated by R&D teams’ supervisors. The four items were used to assessing three aspects of team creativity that are meaningful in R&D settings: newness of idea, significance of idea, and usefulness of idea (Shin and Zhou, 2007). Sample item was ‘This team produces ideas that are novel’.

3.2.5. Control variables

In line with previous research on R&D team creativity, team size, task type and task interdependence were used as controls to partial out their potential influences on team creative efficacy and R&D team creativity (e.g. Leenders et al., 2003; Kratzer et al., 2008; Chen et al., 2010). Task type was measured in five categories according to the products the R&D team develops. Task interdependence was measured using Pearce and Gregersen (1991)’s three-item scale.

3.3. Measurement analyses

We performed a confirmatory factor analysis (CFA) using the LISREL 8.54 computer package to assess the convergent and discriminant validity of our team-level constructs. The CFA yielded an acceptable fit to the data ($\chi^2 = 235.42$, $df = 174$, $CFI = .90$, $IFI = 0.90$, $RMSEA = .065$). We found the factor loading (FL) for each item on its corresponding construct was higher than .50 and there were no cross-loadings between items on any two constructs ($.73 \leq FL \leq 0.84$). As is shown in Table 1, all the Cronbach’s alpha coefficients for the scales exceed 0.7, indicating good internal consistency of the items in the construct. All the composition reliability (CR) values of the latent

variables are greater than .70, and all the average variance extracted (AVE) values are greater than .50 as suggested by Fornell and Larcker (1981), so the convergent validity was satisfied. The root AVE values are significantly greater than the inter-construct correlations, which indicates adequate discriminate validity.

We calculated within-group agreements (r_{wg}) values using James et al.’s (1993) formula. The mean of r_{wg} values for all constructs exceeded the generally accepted 0.70 cutoff for substantial interrater agreement ($.91 \leq \text{mean } r_{wg} \leq 0.98$). In addition, we computed intraclass correlation coefficients using Bliese and Halverson’s (1998) formula. The ICC[1]s exceeded the generally accepted .12 cutoff: external developmental feedback (.22); team creative efficacy (.28); cooperative goal interdependence (.26) and task interdependence (.31). The corresponding ANOVA *F*-statistics were statistically significant ($P < .01$). The ICC[2]s exceeded the .60 cutoff suggested by Glick (1985): external developmental feedback (.61); team creative efficacy (.69); cooperative goal interdependence (.65) and task interdependence (.71). The tests supported the appropriateness of aggregating individual-level data to the team level.

4. Results

Since the results of the structural equation model (SEM) are sensitive to the sample size and the amount of team-level data collected in this study is relatively small ($n_{\text{team-level}} = 84$), it is not appropriate to use SEM to test our hypotheses with LISREL software (Fan et al., 1999). At present, the procedure for examining the conditional indirect effect outlined by Preacher et al. (2007) is a common method for testing moderated mediation with SPSS software. Hence, we used SPSS19.0 software and integrated Baron and Kenny’s (1986) procedure of mediation analysis into Preacher et al. (2007)’s procedure to test our hypotheses. In the first step, we examined the relationships between the independent variable and the dependent variable. The main effect test showed that external developmental feedback was positively related to R&D team creativity ($b = .30$, $P < .05$).

The second step in the analysis involves regressing the mediator (team creative efficacy) on the independent variable (external developmental feedback) and the control variables. The results presented in Table 2 demonstrated that external developmental feedback was significantly and positively related to team creative efficacy ($b = .31$, $P < .01$), providing support

Table 1. Descriptive statistics, composition reliability values, Cronbach's alpha coefficients, correlations, and square roots of AVE¹

Variables	1	2	3	4	5	6	7
1. Team size	n.a.						
2. Task type	−0.12	n.a.					
3. Task interdependence	0.05	0.26*	0.78				
4. External developmental feedback	0.23*	−0.07	−0.04	0.76			
5. Team creative efficacy	0.17	−0.01	−0.15	0.32**	0.79		
6. Cooperative goal interdependence	−0.09	−0.07	−0.04	0.08	−0.01	0.80	
7. R&D team creativity	−0.01	−0.02	−0.04	0.24*	0.39**	0.33**	0.77
Mean	7.01	2.69	4.97	5.07	5.01	5.06	4.52
Median	6.00	3.00	5.11	4.63	5.13	5.30	4.00
SD	3.69	1.35	0.91	1.06	1.09	1.16	1.13
Cronbach's α			0.89	0.80	0.83	0.81	0.85
CR			0.82	0.81	0.84	0.90	0.85

¹ $n = 84$; the root AVE values are marked in bold.* $P < .05$,** $P < .01$.

for the positive effect of external developmental feedback on team creative efficacy. Hypothesis 1 was supported.

In the third step, as the results in Table 2 showed, the previously positive, significant relationship between external developmental feedback and R&D team creativity became weaker significantly when team creative efficacy was added to the equation ($\beta = .30$, $P < .05$ vs. $\beta = .18$, n.s.), combined with the significant effect of team creative efficacy on R&D team creativity ($\beta = .38$, $P < .001$). We concluded that team creative efficacy fully mediated the positive relationship between external developmental feedback and R&D team creativity. Hypothesis 2 was supported.

The fourth step involves a multiple regression predicting the dependent variable (R&D team creativity) from the mediator (team creative efficacy), the moderator (cooperative goal interdependence), the independent variable (external developmental feedback), and the moderator-mediator interaction (cooperative goal interdependence-team creative efficacy). The variables used as components of the interaction term were mean centered. The results in Table 2 indicated a significant effect of the interaction between team creative efficacy and cooperative goal interdependence on R&D team creativity ($b = .28$, $P < .01$), offering support for the moderating effect of cooperative goal interdependence on the relationship of team creative efficacy and R&D team creativity. We plotted this interaction in Figure 2. As shown in Figure 2, team creative efficacy related positively to R&D team creativity only under a high level of

cooperative goal interdependence. Hypothesis 3 was thus confirmed as well.

The last two steps (Step 5 and Step 6) involve using the MODMED method developed by Preacher et al. (2007) to estimate whether the conditional indirect effect of external developmental feedback on R&D team creativity through team creative efficacy differs in strength across low and high levels of cooperative goal interdependence. Step 5 assumes normality of sampling distribution, but the nonparametric Step 6 rests on bootstrapping. The two tests of the conditional indirect effect both indicated that the indirect effect of external developmental feedback on R&D team creativity was significant only when cooperative goal interdependence was high ($Z = 2.55$, $P < .05$). Under low cooperative goal interdependence, the indirect effect of external developmental feedback on R&D team creativity was not significant ($Z = 0.49$, $P > .1$). This furnished support for Hypothesis 4.

5. Discussion

5.1. Theoretical implications and contributions

The results of this study generate several valuable insights with interesting theoretical implications for the R&D team management literature. The first major implication of our findings is that external developmental feedback has a positive indirect effect on R&D team creativity through team creative

Table 2. Test of overall model

Variable	Estimate	SE	<i>t</i>	<i>F</i> value	<i>R</i> ²
Dependent variable model (Step 1): R&D team creativity				1.628	0.076
Team size	−.05	.04	−1.25		
Task type	−.01	.10	−0.08		
Task interdependence	−.03	.14	−0.22		
External developmental feedback	.30	.12	2.46*		
Mediator variable model (Step 2): team creative efficacy				2.752*	0.122
Team size	.01	.03	0.34		
Task type	−.01	.09	−0.09		
Task interdependence	−.16	.13	−1.24		
External developmental feedback	.31	.11	2.75**		
Dependent variable model (Step 3): R&D team creativity				3.796**	0.196
Team size	−.05	.03	−1.47		
Task type	−.01	.09	−0.05		
Task interdependence	.03	.13	0.24		
External developmental feedback	.18	.12	1.49		
Team creative efficacy	.38	.11	3.41***		
Dependent variable model (Step 4): R&D team creativity				7.073***	0.355
Team size	−.04	.03	−1.34		
Task type	.02	.08	0.234		
Task interdependence	−.02	.12	−0.194		
Team creative efficacy	.37	.10	3.78***		
Cooperative goal interdependence	.35	.09	3.92***		
Team creative efficacy × Cooperative goal interdependence	.28	.09	3.01**		
Level of cooperative goal interdependence		Indirect effect	SE	<i>Z</i>	<i>P</i> > <i>Z</i>
Independent variable: external developmental feedback					
Conditional indirect effects assuming normal distribution (Step 5)					
Low cooperative goal interdependence (−1 SD)		.02	.05	0.49	0.62
High cooperative goal interdependence (+1 SD)		.21	.08	2.55*	0.01
Conditional indirect effects with bootstrap method (Step 6)					
Low cooperative goal interdependence (−1 SD)		.02	.05	0.49	0.62
High cooperative goal interdependence (+1 SD)		.21	.08	2.55*	0.01

n = 84.**P* < .05,***P* < .01,****P* < .001.

efficacy. Researchers have argued that social factors may motivate desired group outcomes through increasing collective efficacy (Bandura, 1982) and provided empirical evidence for the motivational effect of contextual factors embedded in the team on team creativity via team creative efficacy (e.g. Shin and Eom, 2014; Boon et al., 2016; Wang et al., 2019). Extending this research, the current study draws on social cognitive theory to identify the

critical role of feedback from external experts, clients, funding sources and other stakeholders outside the organization as the source of motivational effect on R&D team creativity via team creative efficacy. In support of the social cognitive theory, our findings reveal that R&D teams frequently obtaining developmental feedback from the external environment can be more creative because those teams are more confident, and could establish higher exception

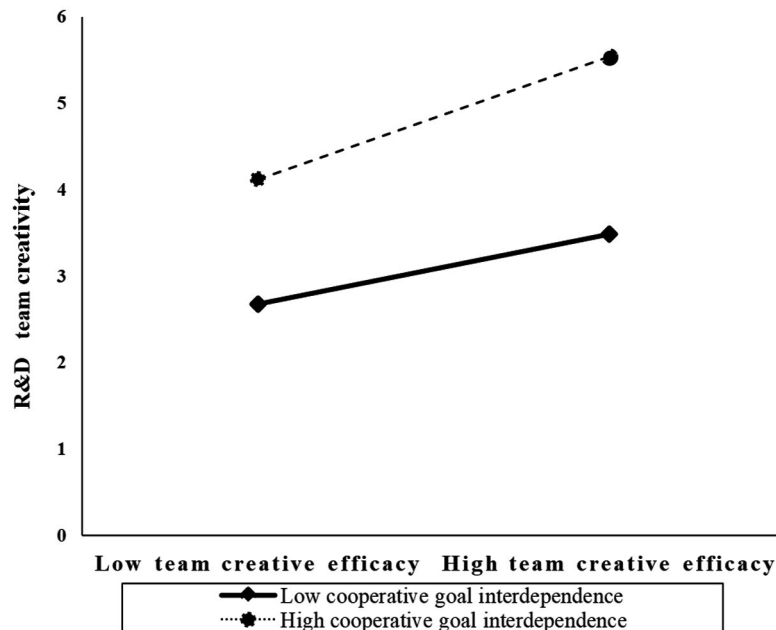


Figure 2. Interaction effect between team creative efficacy and cooperative goal interdependence on R&D team creativity.

about the R&D team's capability to generate creative ideas. The results of this study are consistent with the existing research which demonstrates external communications increase creativity in the R&D team by facilitating the access and processing of information or knowledge (e.g. Tang, 2014, 2016; Hoever et al., 2018). However, this study provides a possible explanation for the positive influence of external communications on R&D team creativity from a new (motivational) perspective. Our results highlight that external communication is not only a source of information, knowledge or other resources for R&D teams but also can cultivate the R&D team's creative efficiency and thus motivate the R&D team creativity.

The second major implication is that cooperative goal interdependence plays a moderating role both in the relationship between team creative efficacy and R&D team creativity and in the indirect effect of external developmental feedback on R&D team creativity via team creative efficacy. Whereas social cognitive theory demonstrates that contextual factors could motivate desired group outcomes via increasing collective efficacy (Bandura, 1982), the theory proposes group members' collective action on the basis of cooperative goals as a necessary condition for the transition from collective efficacy to desired group results (Bandura, 1986). Since little attention has been paid to the potential moderating effect in the relationship between team creative efficacy and team creativity (e.g. Shin and Eom, 2014; Hon and Chan, 2013; Wang et al., 2019), our findings provide a valuable addition to the literature on the motivation

of R&D team creativity in the explication of how team's cooperative goal interdependence strengthens the positive effect of team creative efficacy on team creativity (e.g. Shin and Zhou, 2007; Wang et al., 2016). The results of this study show that under high cooperative goal interdependence, the indirect effect of external developmental feedback on R&D team creativity via team creative efficacy was significant while under low cooperative goal interdependence, that indirect effect was not significant, which adds to previous research by clarifying the role of R&D team's cooperative goal interdependence as the boundary condition of the motivational effect of external communications on R&D team creativity (e.g. Tushman and Katz, 1980; Hoever et al., 2018).

Finally, this study combines the influence of external developmental feedback and internal goal interdependence to establish an integrated theoretical model to examine the social cognitive effect that motivates R&D team creativity. The prior study provides empirical evidence for the influence of external communications and the impact of internal team structures on R&D team's innovation or creativity in separate studies (e.g. Tushman and Katz, 1980; Shin and Zhou, 2007; Chen et al., 2010; García Martínez et al., 2017). However, there have been few concerted efforts to link external communication and internal structure in a unified theoretical framework. As R&D team creativity is often enacted in an open environment using the external and internal complementary innovation resources (Tang, 2016), the current study draws on social cognitive theory to establish

and test a theoretical model in which external developmental feedback acts as the source of the social cognitive effect on R&D team creativity, and internal goal interdependence acts as the boundary condition (motivator) of the above effect. This study thus provides a feasible theoretical framework for use by future researchers as they explore the joint effects that external and internal contextual factors exert on the R&D team's creativity and innovation.

5.2. Implications for management

This study has two practical implications to help organizations more fully motivate R&D team creativity: first, our findings support the value of developmental feedback received from external experts, clients funding sources and other stakeholders outside the organization as a possible way to promote R&D team's creative efficacy and creativity. The R&D team's supervisor should take the initiative to establish external contacts, seek feedback from those contacts, identify and establish a long-term guidance relationship with those who can provide developmental feedback for the improvement of R&D team, so as to enhance team creative efficacy and motivate R&D team creativity through regularly seeking external developmental feedback.

Second, since cooperative goal interdependence could boost the positive indirect effect of external developmental feedback on R&D team creativity, R&D team managers need to make sure that individual goals are compatible with team goals when they recruit R&D team members. They could also inform R&D team members about that compatibility to make a high level of cooperative goal interdependence perceived in the team. In sum, R&D teams should not only actively seek external developmental feedback but also ensure a high level of cooperative goal interdependence in the team to develop a confident, cooperative and creative R&D team.

5.3. Limitations and future directions

The first limitation of this study is that the data may be susceptible to common method variance as we asked R&D team members to self-report their perceived external developmental feedback, team creative efficacy and cooperative goal interdependence. Yet, we conducted a two-phase and two-source data collection with perceived external developmental feedback and team creative efficacy, cooperative goal interdependence collected from R&D team members at different stages, and R&D team creativity collected from supervisors, which can help reduce the threat of common method variance (Podsakoff et al.,

2003). Second, although we used well-established instrument in the previous study to measure R&D team creativity (Shin and Zhou, 2007), the scale is subjective and might induce biases in responses. Objective measurement of R&D team creativity should be developed and used in future empirical studies. Finally, future studies should examine the combined social cognitive effects of external feedback and internal goal interdependence on R&D team creativity with some important variables such as skills diversification and team experience levels controlled for, so as to provide a more comprehensive understanding of the motivational mechanism of R&D team creativity.

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